Solving the nonconvex optimization problem for path planning with obstacle-avoidance constraints

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I. Goals and motivation

I would say my project falls under ”implementing and evaluating known techniques on a specific problem or robotic system”. I want to take a real-life problem that can be framed as a nonconvex optimization problem and try different methods for finding locally optimal solutions. I will talk more about the actual methods I want to use in the ”Technical approach” section. If I have the time, I will compare performance of these methods, which such as the quality of their solution and time to compute the solution.

One idea for a real-life problem I have in mind is path planning when there are obstacles. As discussed in lecture, path planning with obstacles can be framed as a nonconvex optimization problem. I find this problem interesting because it's a problem I always find myself doing. Whether it's walking on campus or driving a car. I am always trying to get somewhere efficiently while navigating around obstacles. It's something I do automatically, so it's reasonable a robot should also be able to do the same. But even though path planning with obstacles is something I can do fairly easily, it's a nonconvex optimization problem and not easy for a robot to do. So I am curious about picking apart a problem like this I usually take for granted.

I also am interested in doing a project about nonconvex optimization because they represent a lot of real world problems, so finding better ways to solve nonconvex optimization problems seems pretty important.

II. Technical approach

Some possible techniques I am thinking of using for finding the shortest path that avoids obstacles are the following:

- Sequential convex programming techniques (like SQP, SCR, Difference of Convex Functions).
- Future material covered in chapter 4 and after.
- Trying relaxation or restriction techniques that work well specifically for the structure of the path planning with obstacle-avoidance problem. (I am not very familiar with the structure of this nonconvex optimization problem yet, so I am not even sure if this can work. However, I am sure others have tried utilizing properties specific to this problem to find optimal solutions, so I will look for any papers that tried this. I will also take a stab at discovering properties myself.)

I also found the following paper titled ”Semi-infinite programming for trajectory optimization with non-convex obstacles” (https://dl.acm.org/doi/10.1177/0278364920983353). The method used here seems different from what's been covered in class, so I am looking deeper into this.

III. Milestones

Assuming I go with path planning with obstacles as my project, these are the following steps I would take.

1) Translate the path planning with obstacles problem into the appropriate nonconvex optimization problem
2) Prepare code to create a model with obstacles and a start to end destination. Have it also take input paths, where the model shows visually the path to be taken. This should behave similar to the code shown in class for this problem.
3) Implementing one of the techniques in the ”Technical approach” section to solve the nonconvex optimization problem.
4) Translate the solution from 3) into the path the robot will take in the model.
5) Depending on my confidence, I will either implement another technique, or test any ideas I have that utilize the structure of the problem.
6) I repeat step 5 until I run out of time.
7) I compare the performances of different approaches I used to solve the problem.
8) I finally begin typing my findings into overleaf.

I am planning my milestones in this way so that I will at least have 1 working implementation of path planning with obstacles as a project (steps 1 - 4). If I stay on schedule, I will try step 5 as well.

IV. Team Information

As of now, I am working alone on this project.

V. Overlap with other work

This project doesn't overlap with anything else I'm doing right now.